

FoodHub — Local Food Delivery app

Final Report

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1. Project Idea

FoodHub is a web application that lets people order food from a local fast-food menu and have it delivered to their door. Instead of calling a restaurant or browsing through several disconnected social media posts to see what is available, users can browse a structured digital menu, search and filter dishes, build a cart, and place an order in a few clicks. The goal of the project is to demonstrate how core HCI principles — clarity, feedback, consistency, and accessibility for users of all ages — can turn a fairly mundane daily task (ordering food) into a fast and pleasant experience.

2. Problem Definition

Many small, local food businesses still rely on phone orders or unstructured social media messages to take orders. This is slow, error-prone, and inaccessible to people who are not comfortable making phone calls or typing out an order manually. There is also no easy way for a customer to track an existing order, search for a dish if they only remember part of its name, or quickly filter the menu by category. On the restaurant side, there is no central dashboard to see incoming orders, update their status, or view basic sales analytics.

3. Proposed Solution

FoodHub provides a complete ordering flow: users browse a categorized menu, add items to a persistent cart, and check out with delivery details that are validated before submission. Registered users can view their past orders. Restaurant staff have a separate, protected Admin area to monitor orders, update order status, and view basic analytics. The application also includes a blog, used to share recipes, kitchen stories, and behind-the-scenes content, with the blog content managed through a headless CMS (Sanity) so non-technical staff can publish new posts without touching code.

4. Target Users

Primary users are individuals who want to order food for delivery — students, working professionals, and families — including less technical users such as older adults, for whom the interface needed to remain forgiving of typos and visually uncluttered. Secondary users are restaurant administrators/staff who need to process and track orders. The application was also explicitly tested on mobile devices (iPhone/Safari), since a large share of food ordering happens on phones.

5. User Personas

User personas were created to better understand the needs of different kinds of FoodHub customers.

Ana Novak — Busy Student

Age: 22 Gender: Female Location: Zagreb, Croatia

Personality: Organized, tech-savvy, always in a rush, prefers quick solutions

Goals & Needs: Order food quickly between classes; Save favorite meals for future orders; Read short blog posts about healthy student meals

Frustrations: Long waiting lines at fast food restaurants; Complicated apps with too many steps before checkout

Context of use: Ana mostly uses her phone to browse the menu and place orders while on campus.

Marko Petrić — Local Restaurant Owner

Age: 40 Gender: Male Location: Split, Croatia

Personality: Hardworking, practical, customer-focused, not very tech-oriented

Goals & Needs: Easily update his menu without coding; Manage incoming orders efficiently

Frustrations: Difficulties managing social media and online tools; Customers calling during rush hours causing confusion

Context of use: Marko uses a computer in his restaurant office to update menu items through the CMS.

Petra Horvat — Young Professional

Age: 28 Gender: Female Location: Rijeka, Croatia

Personality: Efficient, social, enjoys trying new food, values convenience

Goals & Needs: Quickly find nearby lunch options; Use filters (vegetarian, spicy, etc.); Read blog posts about interesting meals

Frustrations: Apps that load slowly or crash; Too many unnecessary ads or notifications

Context of use: Petra uses the web app during work breaks on her smartphone or laptop.

6. Information Architecture

FoodHub's information architecture was shaped by a card sorting exercise that grouped all planned content and functionality into five top-level categories, which then informed the navigation structure and sitemap below.

Category	Content
Home & Discovery	Homepage, Featured Dishes
Menu & Ordering	Menu List, Food Categories, Search/Filter, Item Details, Cart, Checkout
User Account	Login/Register, Profile, Order History, Favorites
Blog & Community	Blog Overview, Post Details, Comments
About & Support	About Us, Contact Form, FAQs, Privacy Policy

6.1 Guest & Customer Role

Guests can browse the Home page, the Menu (with category filters and typo-tolerant search), About, Contact, and the Blog without an account. Adding items to the Cart is also available without logging in, with immediate visual feedback (a pop-up confirmation and an updated cart badge). Checkout and order history, however, require an account, since orders are tied to a specific user.

6.2 Admin Role

Administrators sign in through the same login form, but their account is flagged with an admin role stored in Supabase. This role is checked on the server, inside Next.js middleware, before any /admin route is rendered — a non-admin (or anonymous) visitor is redirected away before the page ever loads. Once inside, administrators see a dedicated dashboard with order statistics, an order management table where order status can be updated, and an analytics page.

6.3 Sitemap

The navigation hierarchy below was defined during the information architecture phase, based on the card sorting categories above:

```
Home (/)
+-- Menu (/menu)
|   +-- Categories (/menu/[category])
|   +-- Product Details (/menu/[slug])
|   +-- Cart (/cart)
|   `-- Checkout (/checkout)
+-- Blog (/blog)
|   `-- Blog Post (/blog/[slug])
+-- User (/user)
|   +-- Login (/login)
|   +-- Register (/register)
```

```
|   +-- Profile (/profile)
|   `-- Order History (/orders)
+-- About (/about)
+-- Contact (/contact)
`-- Legal (/privacy-policy)
```

During implementation, this structure evolved slightly: the menu lives at /book_meal instead of /menu, Profile and Favorites were not implemented in this iteration, and a separate, protected Admin section (/admin — orders management and analytics) and a CMS editor (/studio) were added on top of the original plan to support restaurant staff.

7. Design and User Interface

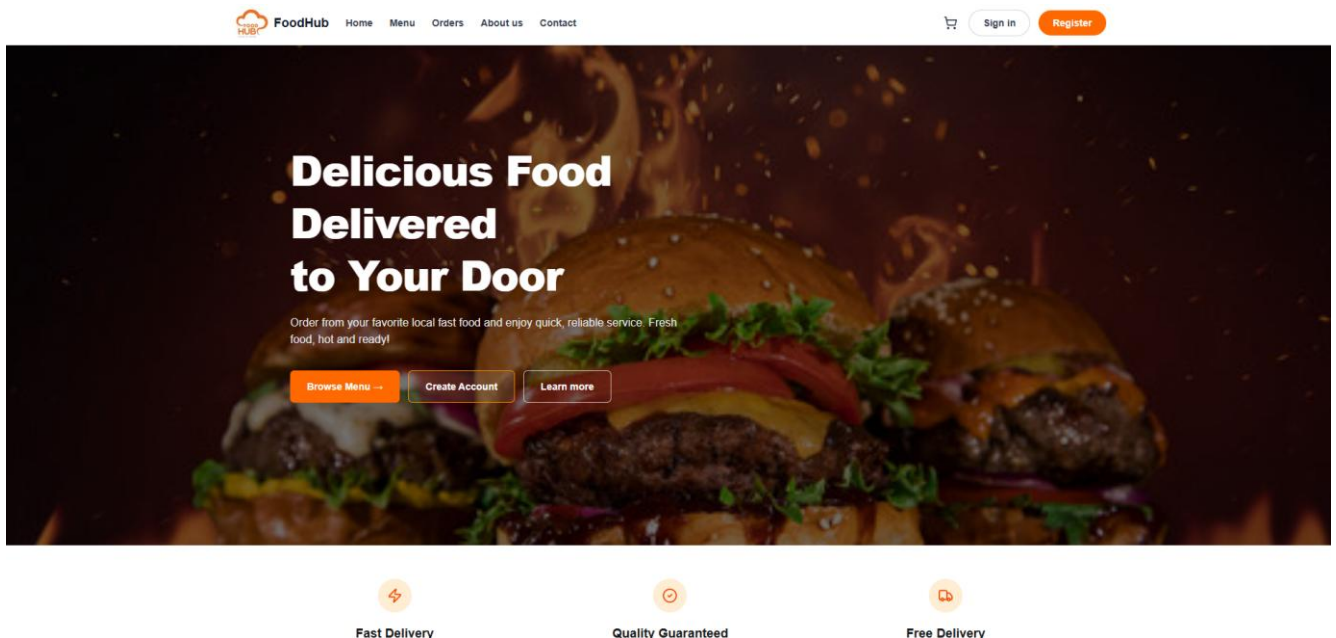
The interface follows a consistent, minimal design language built on an orange accent color over a white/neutral background, with rounded cards, soft shadows, and a fixed top navigation bar present on every public page.

7.1 Home Page

Introduces FoodHub with a hero banner, highlights core benefits (fast delivery, quality, free delivery), shows popular categories, and previews the three latest blog posts.

7.2 Menu Page

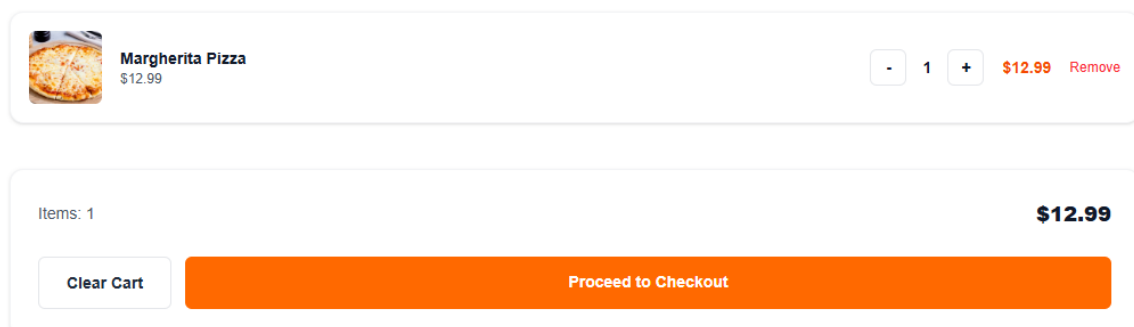
Lists all dishes with a search box and category filter pills (All, Pizza, Burgers, Pasta, Sushi, Salads, Desserts). Search tolerates typos using a Levenshtein-distance match, so guests who are not comfortable typing can still find what they want.



7.3 Cart & Checkout

The cart shows selected items with running totals. Checkout collects name, phone, and delivery address, with inline validation rejecting clearly invalid input (e.g. a name made only of digits) before the order can be submitted.

Your Cart



7.4 Orders Page

Signed-in users see their own past orders with item breakdowns, status badges, and a correctly localized timestamp (Croatia/Zagreb time zone).

FoodHub
My Orders
View previously placed food orders.

Order again

Order #339b48f2
30. lip 2026. 09:17

Items

Margherita Pizza
2 × \$12.99

\$25.98

Delivery details

Name: Nikola Čurak
Phone: +385 916211144
Address: Fra Grge Martića 8
Note: /

7.5 Admin Panel

Protected dashboard showing order totals, revenue, and pending/delivered counts, plus a table to update order status and an analytics breakdown.

FoodHub Admin
Dashboard
Overview of orders, revenue and delivery status.

Total Orders
3

Revenue
\$53.95

Pending
3

Delivered
0

7.6 Login / Registration

Simple, focused forms for signing in or creating an account, validated before submission.

Sign In ×

Email

Password

Sign In

Don't have an account? [Sign up](#)

7.7 Blog

A list of articles mixing text, images, embedded video, and code snippets, with content managed in Sanity Studio rather than hardcoded in the app.

FoodHub Blog

Stories, recipes, and behind-the-scenes

News from the kitchen, ingredient deep-dives, and a peek at how FoodHub is built.



Nov 30, 2025 · Chef Marko

The Secret to Perfect Pizza Dough

Learn the art of making authentic Italian pizza dough from scratch. Our chefs share their tips...



Nov 14, 2025 · Ana Kovač

Farm to Table: Our Ingredient Journey

Discover how we source the freshest local ingredients for our dishes. We work directly...



Sep 27, 2025 · Nikola Ćurak

How We Built FoodHub's Typo-Friendly Search

A behind-the-scenes look at the search feature that helps everyone find their meal, typos and all.



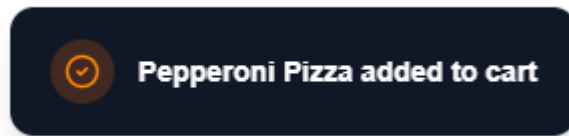
Sep 9, 2025 · Ana Kovač

Behind the Scenes: A Tour of Our Kitchen

8. HCI Principles

Usability

Core tasks — searching the menu, adding to cart, and checking out — require a minimal number of steps. Adding an item to the cart takes a single tap and produces immediate visual confirmation (a toast notification), so the user never has to guess whether the action succeeded.



Consistency

The same navigation bar, card layout, button shapes, and color palette (orange accent, white background, rounded corners) are reused across every page, including the admin panel. Icons are drawn from a single icon set (lucide-react) throughout the app, replacing an earlier mix of emoji and flat icons that a course reviewer flagged as visually inconsistent.



Fast Delivery

Get your food delivered in 30 minutes or less.



Quality Guaranteed

Fresh ingredients and high-quality meals every time.



Free Delivery

Free delivery on orders over a minimum amount.

Popular Categories



Pizza



Burgers



Pasta



Sushi



Salads



Desserts

Learnability

Navigation labels (Home, Menu, Orders, About, Contact) use plain, familiar language. The mobile hamburger menu mirrors the desktop navigation exactly, so switching between devices does not require relearning the structure.

Feedback

The application gives immediate feedback for nearly every action: a toast confirms an item was added to the cart, form fields show inline red error messages when validation fails, and order status badges are color-coded (orange for pending, yellow for preparing, green for delivered).

Error Prevention

Checkout and contact forms validate names (letters only), phone numbers, and email addresses before submission, preventing clearly invalid orders (e.g. a name of “123”) from ever reaching the database. Admin routes are protected at the server level via middleware, preventing accidental or unauthorized access to sensitive order data rather than just hiding a link in the UI.

9. CRAP Principles

Contrast

The orange accent color is reserved for primary actions (“Add to Cart”, “Place order”, active filter pills), making them stand out clearly from secondary, neutral-colored actions like “Back” or “Learn more”.

Repetition

The same dish-card structure (image, title, price, description, action button) repeats across the menu and blog. The same rounded-card / shadow / spacing pattern repeats across every section of every page, building familiarity as the user moves through the app.

Alignment

Content is organized in a consistent grid (1–3 columns depending on screen size) with a shared max-width container, keeping text and cards aligned to the same vertical guides on every page.

Proximity

Related controls are grouped together — for example, the quantity and price of a cart item sit next to each other and separately from the next item, and form labels sit directly above their input fields rather than being separated by unrelated content.

10. Technologies Used

Technology	Purpose in FoodHub
Next.js 16 (App Router)	Frontend framework — file-based routing, server components
TypeScript	Type-safe JavaScript for fewer runtime errors
Tailwind CSS 4	Utility-first responsive styling
Supabase	Authentication, Postgres database, and middleware-based route protection
Sanity.io	Headless CMS for blog content (Studio embedded at /studio)
Web3Forms	Contact form email delivery, no custom backend required
lucide-react	Consistent icon set across the entire app
Vercel	Hosting and deployment

11. Low-Level Theories

Hick–Hyman Law

FoodHub reduces decision time by limiting visible choices: the menu defaults to “All” but offers a small, fixed set of category filters (Pizza, Burgers, Pasta, Sushi, Salads, Desserts) instead of an unstructured list, letting users narrow down options instead of scanning the entire menu at once.

Our Menu

Browse our delicious selection of meals

Q Search for dishes...

All

Pizza

Burgers

Pasta

Sushi

Salads

Desserts

Fitts's Law

Primary actions such as “Add to Cart” and “Place order” are large, full-width buttons placed directly next to the content they act on, minimizing the distance and precision required to hit them — especially important on a touchscreen.



Margherita Pizza

\$12.99

Classic pizza with tomato sauce, mozzarella, and fresh basil.

+ Add to Cart



Pepperoni Pizza

\$14.99

Traditional pizza loaded with pepperoni and cheese.

+ Add to Cart



Classic Burger

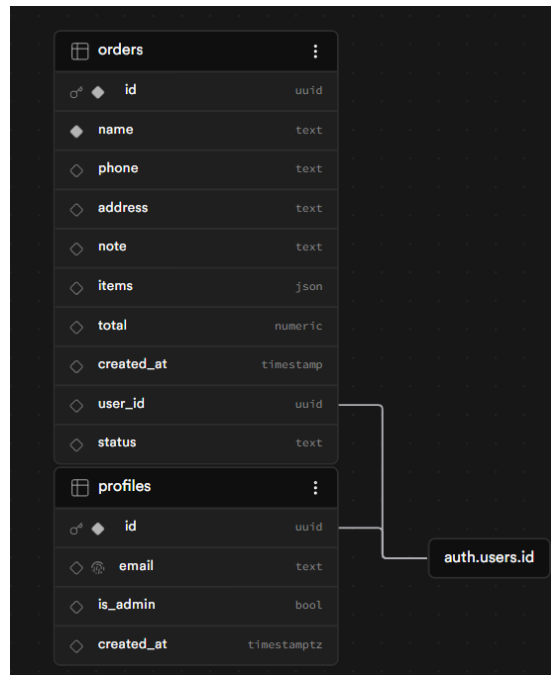
\$10.99

Juicy beef patty with lettuce, tomato, pickles, and sauce.

+ Add to Cart

12. Database and Data Management

FoodHub uses Supabase (PostgreSQL) for authentication and order storage. Users authenticate via Supabase Auth, with session cookies handled through a server-side Supabase client so that Next.js middleware can verify identity and role (customer vs. admin) before a protected page is ever rendered. Orders are stored with the placing user's id, delivery details, ordered items (as a JSON array), total price, status, and a server-generated, ISO-formatted timestamp, which is displayed to the user in the Europe/Zagreb time zone.

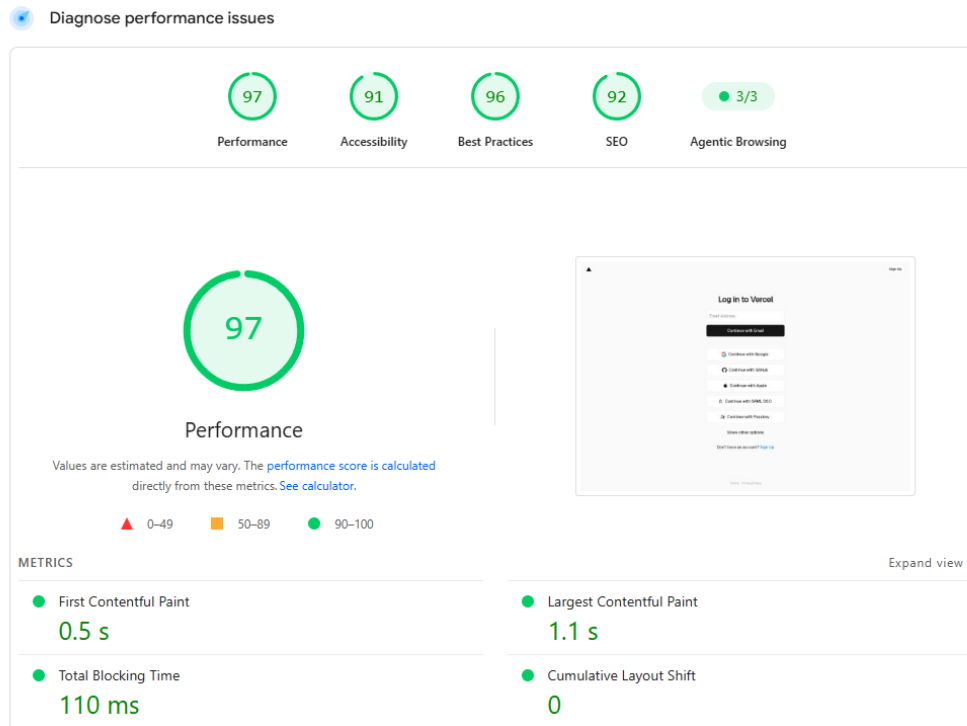


13. Content Management System

Blog content is managed through Sanity, a headless CMS embedded directly in the app at the `/studio` route. A custom schema defines a blog post as a title, slug, author, date, excerpt, cover image, and a flexible content field that mixes rich text paragraphs, images, video embeds (including YouTube links), and code snippets in any order. This means new posts — and changes to existing ones — can be published by editing content in Sanity Studio, without redeploying the application or touching any code.

14. Performance and Optimization

Performance testing was conducted using Google PageSpeed Insights, and the results were excellent. The application demonstrates fast loading times, smooth interactions and overall high performance.



15. Future Improvements

- Real-time order status updates for customers (e.g. via Supabase real-time subscriptions) instead of manual refresh.
- Saved delivery addresses and reorder-in-one-click for returning customers.
- Push/email notifications when an order status changes.
- Ratings and reviews for menu items.
- Expanding the CMS-managed content to menu items, so restaurant staff can update dishes and prices without code changes.

16. Conclusion

FoodHub demonstrates how core HCI and visual design principles — clarity, consistency, forgiving input handling, and immediate feedback — can be applied to a everyday task like food ordering to make it fast and accessible to users of varying technical comfort, from students to older adults. Iterative testing, including real-device mobile testing, surfaced and resolved concrete usability issues (typo-sensitive search, inconsistent iconography, unreachable mobile menu items, broken images on iOS), reinforcing that HCI principles only have value when validated against real user conditions. In its current form, FoodHub fulfills its goal of providing a simple, consistent, and forgiving ordering experience, backed by a content workflow that does not require code changes to update the blog.